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5 Collagen is a component of the wall of the aorta.

(a) (i) Describe the structure of collagen.

(3)

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(ii) Explain the role of collagen in the wall of the aorta.

(2)

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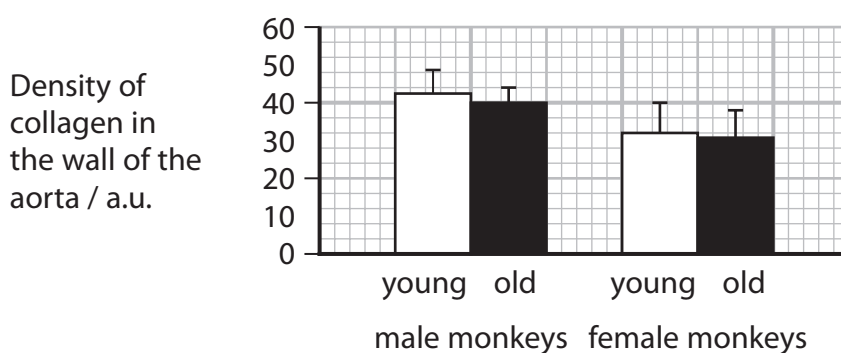
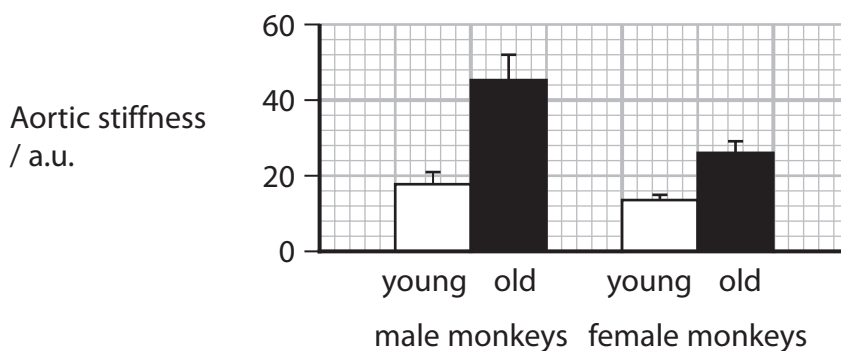


*(b) Age and sex affect the types and density of collagen found in the wall of the aorta.

These are thought to cause aortic stiffness.

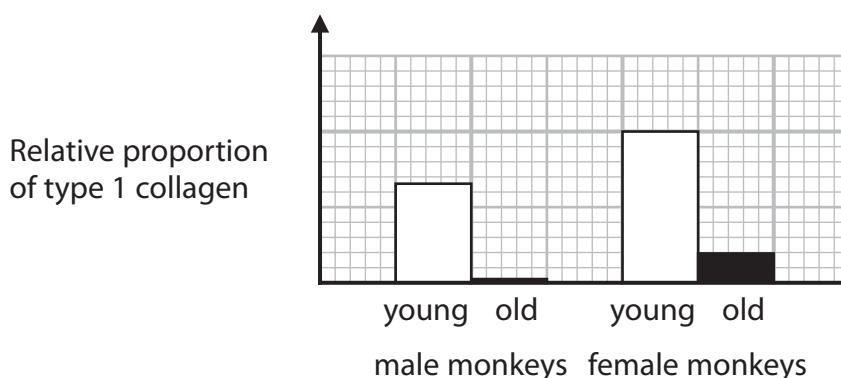
In one study, scientists investigated the effect of age and sex on aortic stiffness and the density of collagen in the wall of the aorta.

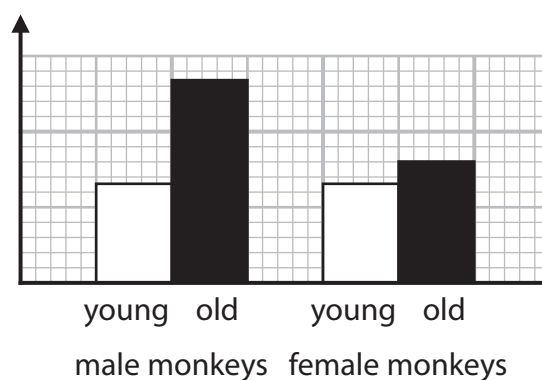
The graphs show the results of this study.



In another study, scientists investigated the effect of age and sex on two types of collagen present in the wall of the aorta.

The graphs show the relative proportion of each type of collagen in each group of monkeys.





Comment on the effect of age and sex on the types of collagen found in the wall of the aorta and aortic stiffness.

Use the information in the graphs to support your answer.

(6)

(Total for Question 5 = 11 marks)

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